

Sales & Order Management Analytics

SQL + Power BI Portfolio Project Documentation

Project Type: Data Analytics / Business Intelligence

Tools: MySQL, Power BI

Core Data: Customers, Products, Orders, Order Items, Payments

Purpose

This project analyzes sales and order-management data to understand revenue performance, orders, products, customers, geography, payments, and operational status. SQL is used for data preparation and business analysis, while Power BI is used to present the results in an interactive dashboard.

1. Business Problem

A business needs a clear view of how its orders and sales are performing. Raw transactional data can make it difficult to identify the most valuable products and customers, understand monthly sales patterns, and distinguish completed, pending, and cancelled orders.

2. Project Objectives

- Calculate important sales and order KPIs.
- Analyze monthly sales trends.
- Identify top-performing products and categories.
- Evaluate customer and city-level performance.
- Analyze repeat customers and order status.
- Review payment performance.
- Build an interactive Power BI dashboard for decision-making.

3. Database Structure

Table	Purpose
customers	Customer details and customer-level analysis
products	Product, category, price and inventory information
orders	Order date, customer, status and order amount
order_items	Products and quantities included in each order
payments	Payment amount, date and payment status

4. SQL Analysis

The SQL layer contains database setup, table creation, data loading, validation, KPI calculations, business analysis queries and reusable views. The analysis covers completed revenue, completed orders, average order value, monthly sales, category sales, top products, customer performance, city performance, repeat customers, order status, payments and inventory.

Example business questions answered

- What is the total completed sales revenue?
- How many completed orders are there?
- What are the top 10 products?
- Which categories generate the most sales?
- Which customers contribute the most revenue?
- How does sales performance change by month?
- How many customers make repeat purchases?
- What is the distribution of order and payment statuses?

5. Power BI Dashboard

Power BI is used as the reporting layer. The dashboard converts SQL-backed transactional data into visual KPIs and charts so that users can quickly understand sales performance.

Recommended dashboard sections

Dashboard Area	Purpose
KPI Cards	Total Sales, Total Orders and other headline metrics
Sales Trend	Track monthly sales performance
Category Analysis	Compare sales across product categories
Product Analysis	Highlight top-performing products
Customer Analysis	Understand customer contribution
Geographic Analysis	Compare sales by city
Filters / Slicers	Interact with the report by relevant dimensions

6. Key Business Insights

The completed analysis is designed to help a business identify revenue drivers, monitor order health, recognize high-value products and customers, compare geographic performance, and understand payment issues. These insights can support inventory planning, customer targeting, sales strategy and operational improvements.

7. Project Deliverables

The recommended GitHub repository structure is:

```
Sales-Order-Management-Analytics/  
■■■ SQL/  
■ ■■■ Sales_Order_Management_Analytics_COMPLETE.sql  
■■■ PowerBI/  
■ ■■■ Sales_Order_Analytics.pbix  
■■■ Screenshots/  
■ ■■■ Dashboard_Page1.png  
■ ■■■ Dashboard_Page2.png  
■■■ README.md
```

8. Conclusion

This project demonstrates an end-to-end analytics workflow: transactional data is structured and analyzed using SQL, then transformed into an interactive Power BI report. It is suitable as a portfolio project because it demonstrates database knowledge, SQL analysis, KPI development, data visualization and business-oriented thinking.

Portfolio note: Add the final Power BI screenshots and GitHub repository link before publishing the project on LinkedIn.